

ADP services does not exactly match the distribution of firms across the broader US economy. Further details on differences in firm composition can be found in [Cajner et al. \(2018\)](#) and [ADP Research \(2025\)](#).¹³

3.2 Occupational AI Exposure

We use two approaches for measuring occupational exposure to AI. The first uses exposure measures from [Eloundou et al. \(2024\)](#), who estimate AI exposure by O*NET task using ChatGPT validated with human labeling. They construct occupational exposure measures by aggregating the task data to the 2018 SOC code level. We focus on the GPT-4 based β exposure measures from their paper.

The second approach we take uses generative AI usage data from the Anthropic Economic Index ([Handa et al., 2025](#)). This index reports the estimated share of queries pertaining to each O*NET task based on several million conversations with Claude, Anthropic’s generative AI model. It aggregates the data to the occupational level based on these task shares. One feature of the Anthropic Economic Index is that for each task it reports estimates of the share of queries pertaining to that task that are “automotive,” “augmentative,” or none of the above. We use this as an estimate of whether AI usage for an occupation is primarily complementary or substitutable with labor.¹⁴

We use a 2010 SOC code to 2018 SOC code crosswalk from the BLS to merge the exposure measures to the payroll data. Table [A1](#) shows example occupations for each AI exposure measure.

3.3 Other Data

To compare employment changes for teleworkable versus non-teleworkable occupations, we use data from [Dingel and Neiman \(2020\)](#). We use the Personal Consumption Expenditure index from the BLS to compute real earnings, indexed to October 2017. We use monthly Current Population Survey (CPS) data as a comparison for our main findings.

¹³[Cajner et al. \(2018\)](#) find a somewhat higher share of manufacturing and services firms compared to the QCEW using data from March 2016. They also find that ADP somewhat overrepresents firms in the Northeast. In addition, firms using ADP tend to grow faster on average than the typical firm in the US economy.

¹⁴[Handa et al. \(2025\)](#) use Claude to classify conversations into six categories. Directive and Feedback Loop conversations are considered Automotive; Task Iteration, Learning, and Validation are considered Augmentative. They also instruct the model to choose None of the above “liberally.” Table [A2](#) reproduces Table 1 from [Handa et al. \(2025\)](#) and shows more details about the automation and augmentation measures.

4 Results

4.1 Fact 1: Employment for young workers has declined in AI-exposed occupations

Consider software engineers and customer service agents, two occupations frequently considered to be highly exposed to generative AI tools. Media attention has raised the specter of widespread employment disruption for young software engineers (Thompson, 2025; Raman, 2025; Allen, 2025; Horowitch, 2025).

Figure 1 shows employment changes by age group for these occupations, normalized to 1 in October 2022. While the types of work and workers in each of these occupations differs in many ways, both occupations present a similar pattern: employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow. By September 2025, employment for software developers aged 22-25 declined nearly 20% compared to its peak in late 2022. Figure A1 shows that a similar pattern holds for computer occupations and service clerks more generally.

Figure A2 shows four other professions as case studies, spanning varying levels of AI exposure according to Eloundou et al. (2024). Marketing and sales managers, in the fourth quintile of AI exposure, show a decline in employment for young workers much like the case of software and customer service, albeit with smaller magnitudes. Front-line production and operations supervisors, in quintile 3, show an increase in employment for young workers, though the growth in employment is smaller than the increase for workers over age 35.

In contrast, the trends for occupations that Eloundou et al. (2024) rated as less exposed do not fit the pattern of the more exposed occupations. Stock clerks and order fillers, in quintile 2, show no obvious difference by age. Strikingly, the series for health aides, comprising nursing aides, psychiatric aides, and home health aides, show a quite different trend from software or customer service: employment for young workers has been growing *faster* than for older workers.

Figure 2 shows these patterns hold more generally across professions. The top left plot shows a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment. For older age groups, we find